ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 2 - Due date 01/27/26

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Submission Instructions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A02_Sp26.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages

R packages needed for this assignment:"forecast","tseries", and "dplyr". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(readxl)
```

Data set information

Consider the data provided in the spreadsheet "Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source on our **Data** folder. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. The spreadsheet is ready to be used. Refer to the file "M2_ImportingData_XLSX.Rmd" in our Lessons folder for instructions on how to read *.xlsx* files.

```r
#Importing data set
df_energy <- read_xlsx("../Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
```

Question 1

You will work only with the following columns: Total Biomass Energy Production, Total Renewable Energy Production, Hydroelectric Power Consumption. Create a data frame structure with these three time series only. Use the command head() to verify your data.

```r
df_select <- df_energy %>%
  select(`Total Biomass Energy Production`,
         `Total Renewable Energy Production`,
         `Hydroelectric Power Consumption`)
head(df_select)
```

```
## # A tibble: 6 x 3
##   Total Biomass Energy Productio~1 Total Renewable Ener~2 Hydroelectric Power ~3
##   <chr>                            <chr>                  <chr>
## 1 (Trillion Btu)                   (Trillion Btu)         (Trillion Btu)
## 2 129.787                          219.839                89.562
## 3 117.338                          197.33                 79.544
## 4 129.938                          218.686                88.284
## 5 125.636                          209.33                 83.152
## 6 129.834                          215.982                85.643
## # i abbreviated names: 1: ‘Total Biomass Energy Production‘,
## #   2: ‘Total Renewable Energy Production‘,
## #   3: ‘Hydroelectric Power Consumption‘
```

Question 2

Transform your data frame in a time series object and specify the starting point and frequency of the time series using the function ts().

```r
ts_energy <- ts(df_select, start=c(1973, 1), frequency=12)
```

Question 3

Compute mean and standard deviation for these three series.

```r
colMeans(ts_energy)
```

```
##   Total Biomass Energy Production Total Renewable Energy Production
##                          315.9511                          316.7334
##   Hydroelectric Power Consumption
##                          315.1120
```

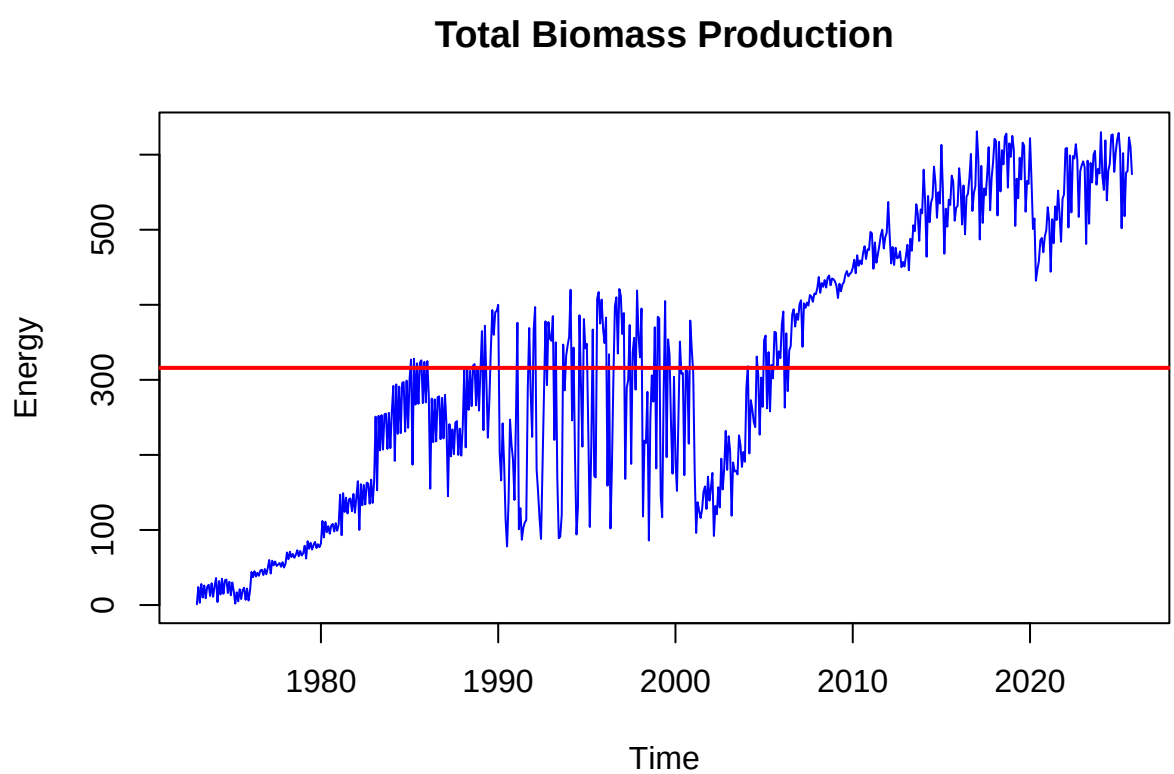
```r
apply(ts_energy, 2, sd)
```

```
##   Total Biomass Energy Production Total Renewable Energy Production
##                          182.5330                          182.6499
##   Hydroelectric Power Consumption
##                          181.2336
```

Question 4

Display and interpret the time series plot for each of these variables. Try to make your plot as informative as possible by writing titles, labels, etc. For each plot add a horizontal line at the mean of each series in a different color.

```r
plot(ts_energy[,1], main="Total Biomass Production", ylab="Energy", col="blue")
abline(h=mean(ts_energy[,1]), col="red", lwd=2)
```



Question 5

Compute the correlation between these three series. Are they significantly correlated? Explain your answer.

```r
energy_correlation <- cor(ts_energy)
print(energy_correlation)
```

```
##                                   Total Biomass Energy Production
## Total Biomass Energy Production                        1.0000000
## Total Renewable Energy Production                      0.9687978
## Hydroelectric Power Consumption                       -0.1255177
##                                   Total Renewable Energy Production
## Total Biomass Energy Production                         0.96879782
## Total Renewable Energy Production                       1.00000000
## Hydroelectric Power Consumption                        -0.07325884
##                                   Hydroelectric Power Consumption
## Total Biomass Energy Production                       -0.12551766
## Total Renewable Energy Production                     -0.07325884
## Hydroelectric Power Consumption                        1.00000000
```

Yes, there is a highly significant positive correlation between Total Biomass and Total Renewable Energy Production (r is approximately 0.97). This is because biomass is a major component of renewable energy, and both have followed a similar upward trend over time. However, Hydroelectric Power Consumption is not significantly correlated with the other two series (absolute r<0.13). This indicates that hydroelectric output is likely driven by independent factors such as hydrological conditions (rainfall/seasonality) rather than the general industrial growth trend observed in biomass and total renewables.

Question 6

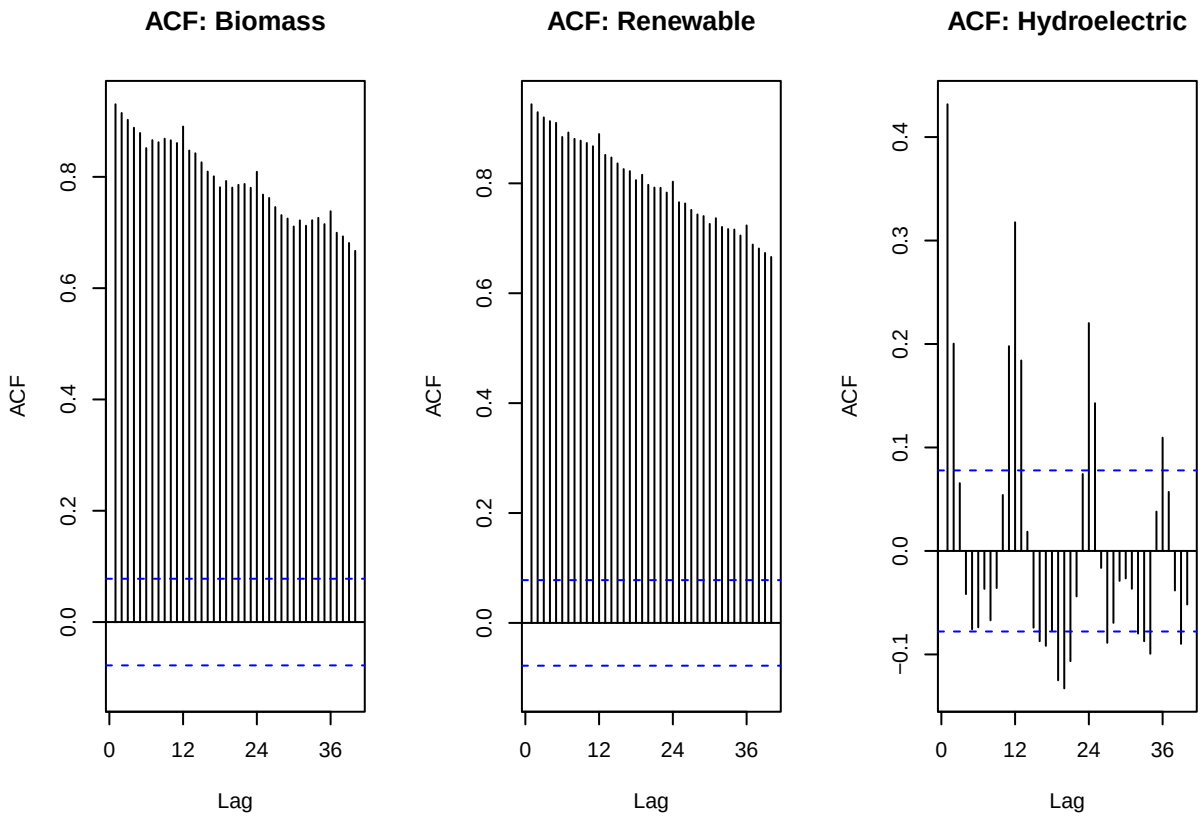
Compute the autocorrelation function from lag 1 up to lag 40 for these three variables. What can you say about these plots? Do the three of them have the same behavior?

```r
par(mfrow=c(1,3))

# 1. Total Biomass
Acf(ts_energy[,1], lag.max=40, main="ACF: Biomass")

# 2. Total Renewable
Acf(ts_energy[,2], lag.max=40, main="ACF: Renewable")

# 3. Hydroelectric
Acf(ts_energy[,3], lag.max=40, main="ACF: Hydroelectric")
```



The ACF plots reveal that the three variables do not exhibit the same behavior:

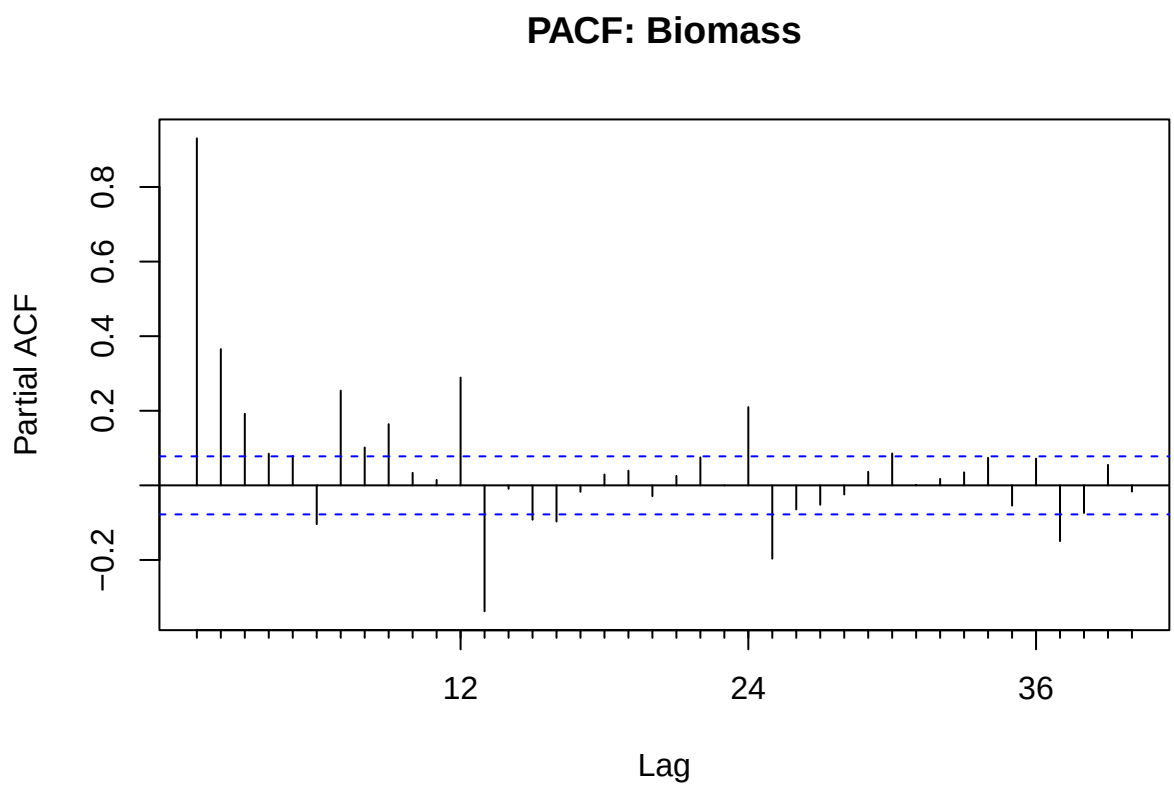
Total Biomass & Total Renewable Energy: These two series show a very similar pattern where the ACF values decrease very slowly and linearly as the lags increase. This is a classic sign of a strong trend (non-stationarity) in the data, indicating that past values have a long-lasting influence on future values.

Hydroelectric Power Consumption: This series shows a distinctly different behavior. The ACF plot exhibits a wave-like or sinusoidal pattern that repeats every 12 lags. This indicates a dominant seasonal component, where the data repeats a specific pattern every year due to cyclical factors like weather or precipitation.

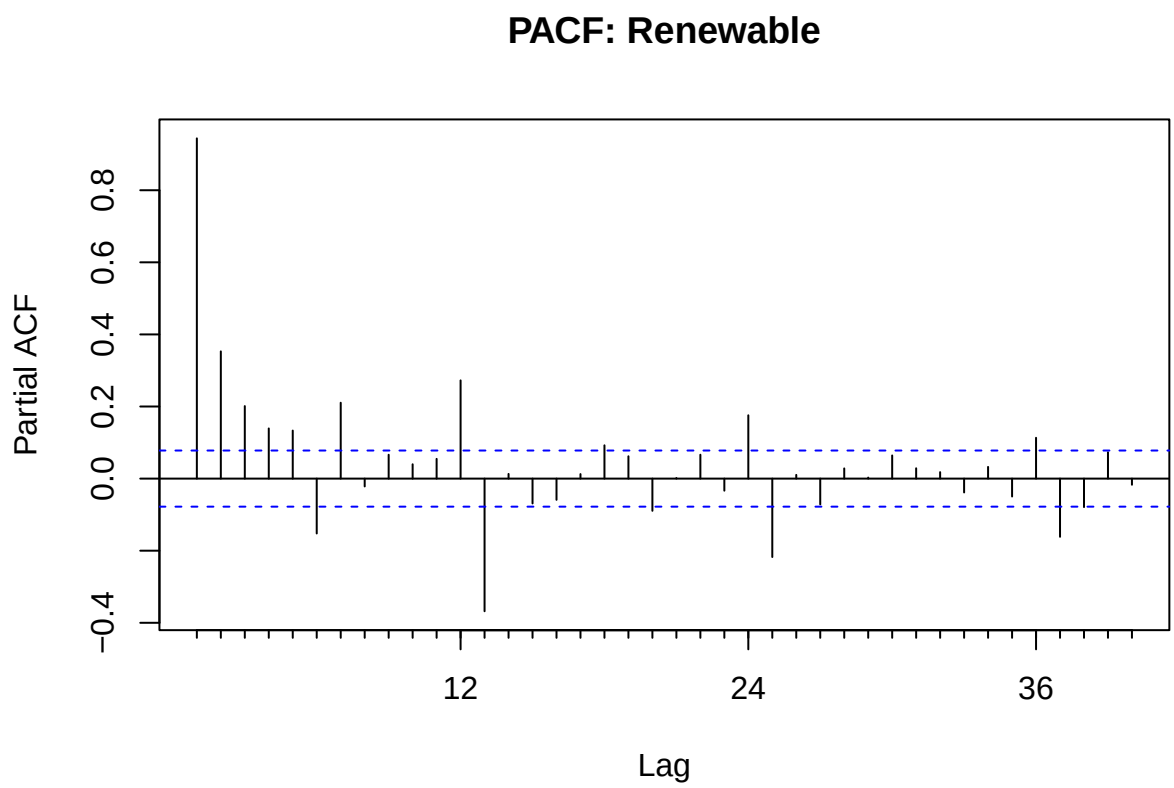
Question 7

Compute the partial autocorrelation function from lag 1 to lag 40 for these three variables. How these plots differ from the ones in Q6?

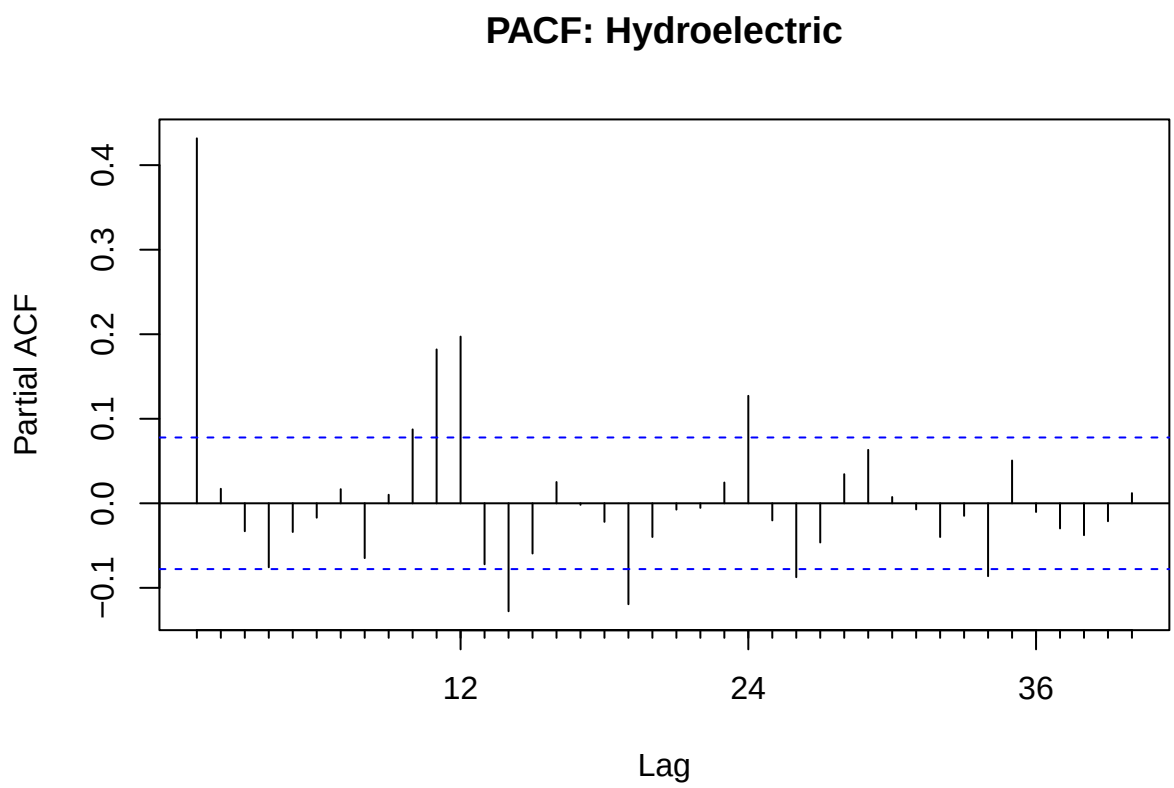
```
Pacf(ts_energy[,1], lag.max=40, main="PACF: Biomass")
```




```
Pacf(ts_energy[,2], lag.max=40, main="PACF: Renewable")
```




```
Pacf(ts_energy[,3], lag.max=40, main="PACF: Hydroelectric")
```



The PACF plots exhibit a much more rapid decay in correlation compared to the ACF plots in Q6, allowing for a clearer identification of the underlying process:

Total Biomass & Total Renewable Energy: While the ACF in Q6 showed a slow, linear decline due to a strong trend , the PACF shows a very high and significant spike at Lag 1, followed by an abrupt drop.

Identification of Seasonality: Crucially, unlike the ACF where it was less obvious, the PACF for these two variables also shows significant spikes at Lag 12 and Lag 24 that exceed the confidence intervals (blue dashed lines). This indicates that while the trend is dominant, there is also a secondary seasonal component present in biomass and renewable production.

Hydroelectric Power Consumption: This series continues to show very prominent and significant spikes at Lag 12, 24, and 36 in the PACF. This confirms that even after controlling for intermediate months, the seasonal cycle remains the strongest driver for hydroelectric data.

Summary of Difference: Q6 (ACF) caught the cumulative effect of the trend and seasonality, making the plots for Biomass and Renewable look very "dense." Q7 (PACF) removes those intermediate correlations, revealing that the current values are most directly influenced by the immediate previous month (Lag 1) and the same month from previous years (Lag 12, 24).